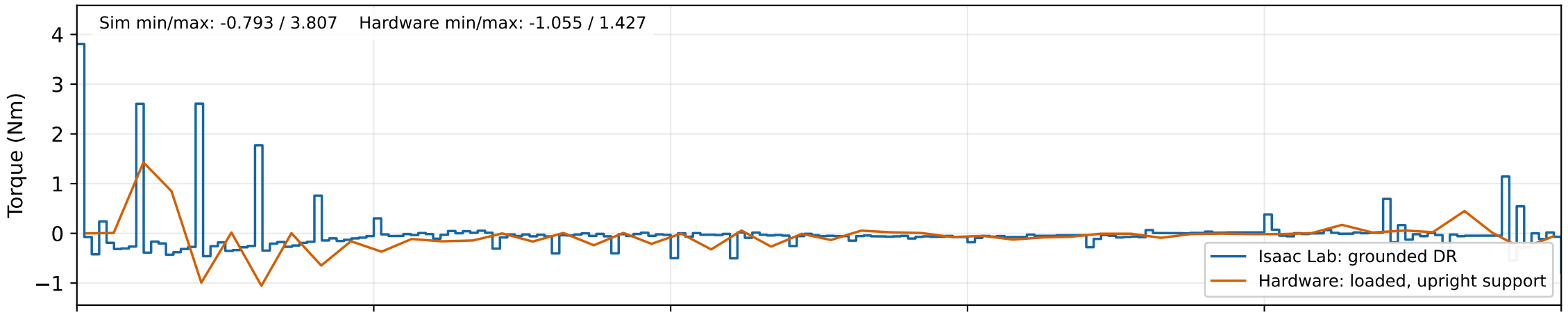
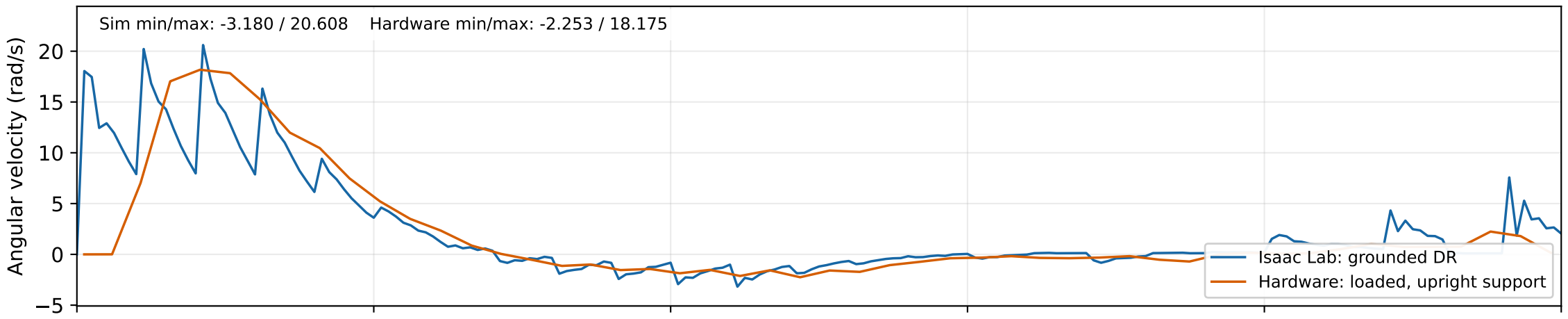
# Loaded steering replay versus Isaac Lab

Model_2500 · source episode 0–5 s · hardware MIT packets: nominal 200 Hz
Recorded Isaac evaluation: 400 Hz physics, 50 Hz targets · all 250 requested targets match
Actual clocks retained: hardware reaches the source-window end at 5.225 s; no time warp

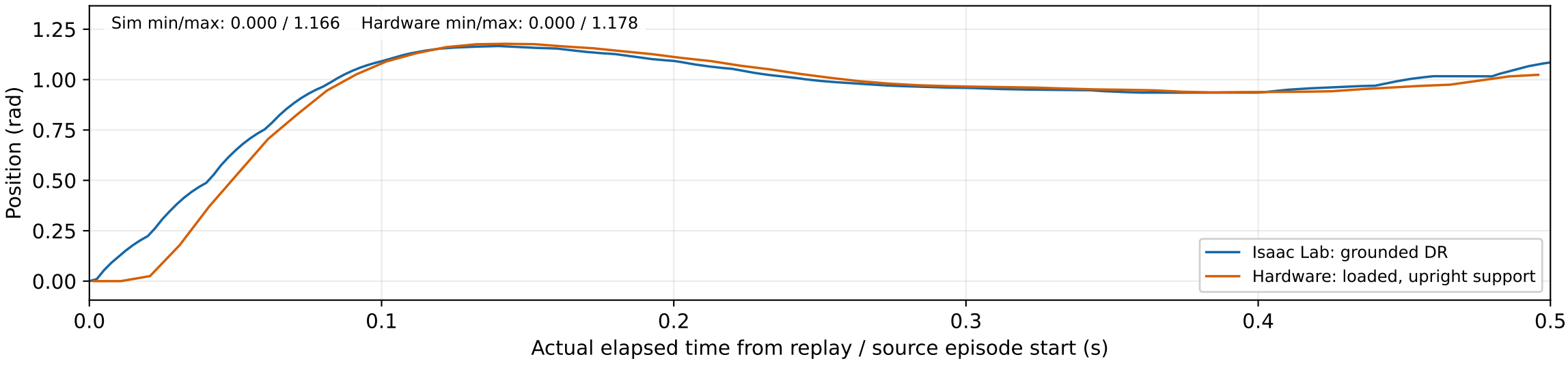
## Actuator torque: simulation effort versus hardware A1 current × Kt




## Steering angular velocity




## Steering position from each run’s fixed straight zero




Hardware: tyre carries normal load, frame supported upright; Kp/Kd = 10/0.2.
Simulation: grounded DR bike balancing freely; Kp/Kd = 10.6517/0.209184. Support and loads differ.
Torque is before passive losses / current-derived, not net shaft torque. No smoothing, fitted delay or bias correction.